

算法 (AI) 第二周优秀作业

第一题

$$\begin{aligned}T(n) &= 2T(n/2) + n \log n \\&= 4T(n/4) + n \log \frac{n}{2} + n \log n \\&= 8T(n/8) + n \log \frac{n}{4} + n \log \frac{n}{2} + n \log n \\&= 16T(n/16) + n \log \frac{n}{8} + n \log \frac{n}{4} + n \log \frac{n}{2} + n \log n \\&\uparrow n = 2^k \cdot 2^1 \\T(n) &= nT(1) + n \left(\log \frac{n^k}{2^0 \cdot 2^1 \cdots 2^{k-1}} \right) \\&= nT(1) + n \log \frac{n^k}{2^{k(k-1)/2}} \\&= nT(1) + n \log n^k - n \log 2^{k(k-1)/2} \\&= nT(1) + nk \log n - n \frac{k(k-1)}{2} \\&= n + n \log n \log n - \frac{1}{2} n \log n (\log n - 1) \\&= n + \frac{1}{2} n \log n \log n + \frac{1}{2} n \log n \\&= O(n \log n \log n)\end{aligned}$$

第二题

2. 解:

$$(1) \quad 3n^2 + 10n = O(n^2)$$

$$(2) \quad n^2/10 + 2^n = O(2^n)$$

$$(3) \quad 21 + 1/n = O(1)$$

$$(4) \quad \log n^3 = O(\log n)$$

$$(5) \quad 10/\log 3^n = 10n \log 3 = O(n).$$

第三题

3. 答: 由大O定义可知 $O(1) = O(2)$, 用 $O(1)$ 和 $O(2)$ 表示同一个函数时, 差别仅在于常数因子 C .

第四题

$$T_{avg}(N) = \Theta(f(n))$$

$$= \sum_{I \in D_N} p(I) T(N, I)$$

$$\leq \sum_{I \in D_N} p(I) \max_{I' \in D_N} T(N, I')$$

$$= T(N, I') \sum_{I \in D_N} p(I)$$

$$= T(N, I')$$

$$= \cancel{T_{max}(N, I')} T_{max}(N)$$

$$T_{avg}(N) = \Theta(f(n)) \leq T_{max}(N, I')$$

$$T_{max}(N) = \Omega(T_{avg}(N)) = \Omega(\Theta(f(n)))$$

$$= \Omega(f(n))$$